

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	25 June 2026
Team ID	Not Given
Project Name	Rising Waters
Maximum Marks	4 Marks

Team Members	Skill Wallet Team ID's	Roll Numbers (College: ACET)
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Brainstorm & Idea Prioritization Template:

During brainstorming, several disaster management ideas were evaluated based on feasibility, impact, accuracy, scalability, and implementation complexity. A machine learning-based flood prediction system was prioritized because it addresses a critical real-world problem, provides timely flood risk predictions using weather data, supports disaster preparedness, and can be easily deployed through a Flask web application for practical use.

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Brainstorm & idea prioritization for Rising Waters

Use this brainstorming template to identify, evaluate, and prioritize innovative ideas for developing a machine learning-based flood prediction system. The goal is to improve early flood warnings, support disaster management, and enable timely decision-making using historical weather data.

- 🕒 10 minutes to prepare
- 🕒 1 hour to collaborate
- 👥 2-8 people recommended

Before you collaborate

Effective planning ensures productive brainstorming. Review the project objectives, available flood datasets, machine learning algorithms, and deployment requirements before generating ideas for the solution.

🕒 10 minutes

A Team gathering

Define team roles such as dataset preparation, machine learning model development, Flask application development, testing, and documentation. Share relevant research papers and project requirements before the session.

B Set the goal

Identify the primary objective: to develop an accurate and scalable flood prediction system that provides early warnings using historical weather parameters and machine learning.

C Learn how to use the facilitation tools

Use the Facilitation Superpowers to run a happy and compare algorithms including Decision Tree, Random Forest, KNN, and XGBoost.

[Open article](#) →

1 Define your problem statement

Develop a reliable machine learning-based flood prediction system that accurately predicts flood risk using weather parameters such as annual rainfall, cloud visibility, humidity, temperature, and seasonal rainfall, enabling authorities to make timely decisions and reduce disaster impacts.

🕒 5 minutes

PROBLEM

How might we build an accurate, scalable, and user-friendly flood prediction system that provides early flood warnings using machine learning and historical weather data to help authorities reduce flood-related risks and improve disaster preparedness?

Key rules of brainstorming

To conduct an effective brainstorming session:

- 😊 Stay focused on improving flood prediction accuracy.
- 💡 Encourage innovative ideas and alternative solutions.
- 👂 Respect every team member's suggestions.
- 👂 Evaluate ideas based on feasibility and real-world impact.
- 🗨️ Prioritize scalable and deployable solutions.
- 👁️ If possible, be visual. Support decisions using data, research, and model performance.

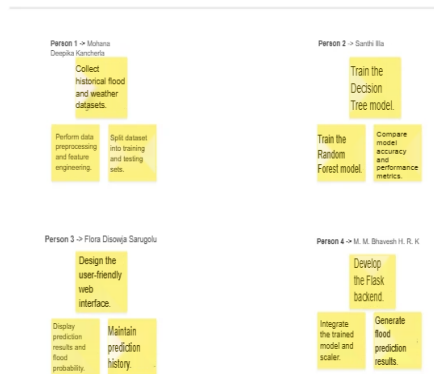
Step-2: Brainstorm, Idea Listing and Grouping

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Brainstorm

Collect historical flood and weather datasets from reliable sources. Perform data preprocessing by handling missing values and scaling features. Compare Decision Tree, Random Forest, KNN, and XGBoost algorithms. Select the model with the highest prediction accuracy. Develop a Flask web application for flood prediction. Validate weather inputs before prediction. Display Flood / No Flood prediction with probability. Store prediction history for future analysis. Test the application using historical weather data. Deploy the solution on IBM Cloud for public access.

20 minutes



3

Group ideas

Model & Accuracy
Compare Decision Tree, Random Forest, KNN, and XGBoost. Evaluate using Accuracy, Precision, Recall, and F1-Score. Select the highest-performing model. Save the trained model and scaler.

User Interface
Flask web application, weather input form, flood prediction dashboard, Prediction history, User-friendly interface.

Data Processing
Collect historical flood datasets, Data cleaning and preprocessing, Feature engineering, Train/Test data split, Feature scaling.

20 minutes

TIP
Add customizer tags to sticky notes to relate a topic to the domain, objective, and compare historical ideas to themes within your mind.

Deployment & Validation

Validate user inputs, Handle invalid or missing data, Test with historical weather records, Deploy on IBM Cloud, Enable real-time flood prediction access.

\Step-3: Idea Prioritization

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Prioritize

High Importance + High Feasibility

Develop the flood prediction model using XGBoost for high prediction accuracy. Collect and preprocess historical weather and flood datasets. Compare Decision Tree, Random Forest, KNN, and XGBoost models. Integrate the best-performing model into a Flask web application. Display Flood / No Flood prediction with probability. Validate user inputs and provide real-time prediction results. Deploy the application on IBM Cloud for public accessibility.

High Importance + Low Feasibility

Integrate real-time weather API for live flood prediction. Support prediction for multiple districts simultaneously. Generate automated SMS/Email alerts for authorities. Implement GIS-based flood risk visualization. Enable continuous model retraining using new weather data.

Low Importance + High Feasibility

Improve the user interface with responsive design. Store prediction history in the database. Add prediction graphs and statistical reports. Include help/documentation and user guidance. Improve application performance and loading speed.

Low Importance + Low Feasibility

Develop Android and iOS mobile applications. Integrate IoT flood sensors for live monitoring. Add multilingual user interface support. Predict floods several days in advance using satellite imagery. Expand the system to support other natural disasters (landslides, cyclones, droughts).

TIP
Participants can use their cursors to point at where sticky notes should go on the grid. The facilitator can confirm the spot by using the **Enter** key on the keyboard.

20 minutes

